CRISPY CRUST ANALYTICS - EMPOWERING BUSINESS

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CRISPY CRUST ANALYTICS

How can Crispy Crust Corner (a hypothetical pizza company) implement a data analytics solution that is both robust and scalable, capable of efficiently processing a large volume of data to derive valuable insights such as trends, ordering behaviors, menu performance, customer preferences, and essential business metrics?

SOLUTION OVERVIEW

In assisting Crispy Crust Corner in leveraging their sales data for actionable insights, we propose a comprehensive approach utilizing SQL with an ETL (Extract, Transform, Load) methodology.

We will start by uploading all four datasets to MySQL and consolidating them into a single dataset containing all the necessary variables for analysis.

Leveraging SQL's capabilities, we will uncover key metrics such as total revenue, average order size, best/worst selling menu items etc.

This approach empowers stakeholders to gain actionable insights into sales trends, order patterns, and menu performance, facilitating informed decision-making.

DATASETS INSIGHT

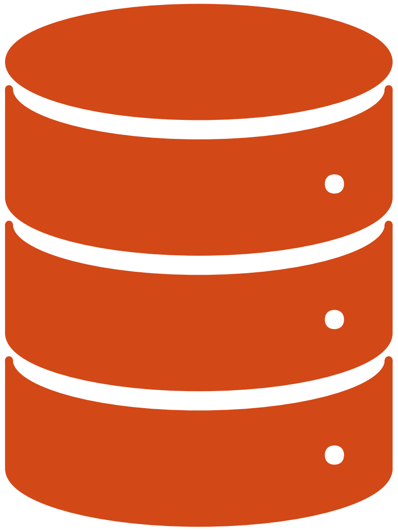
Dataset : Total 4 datasets (csv files)

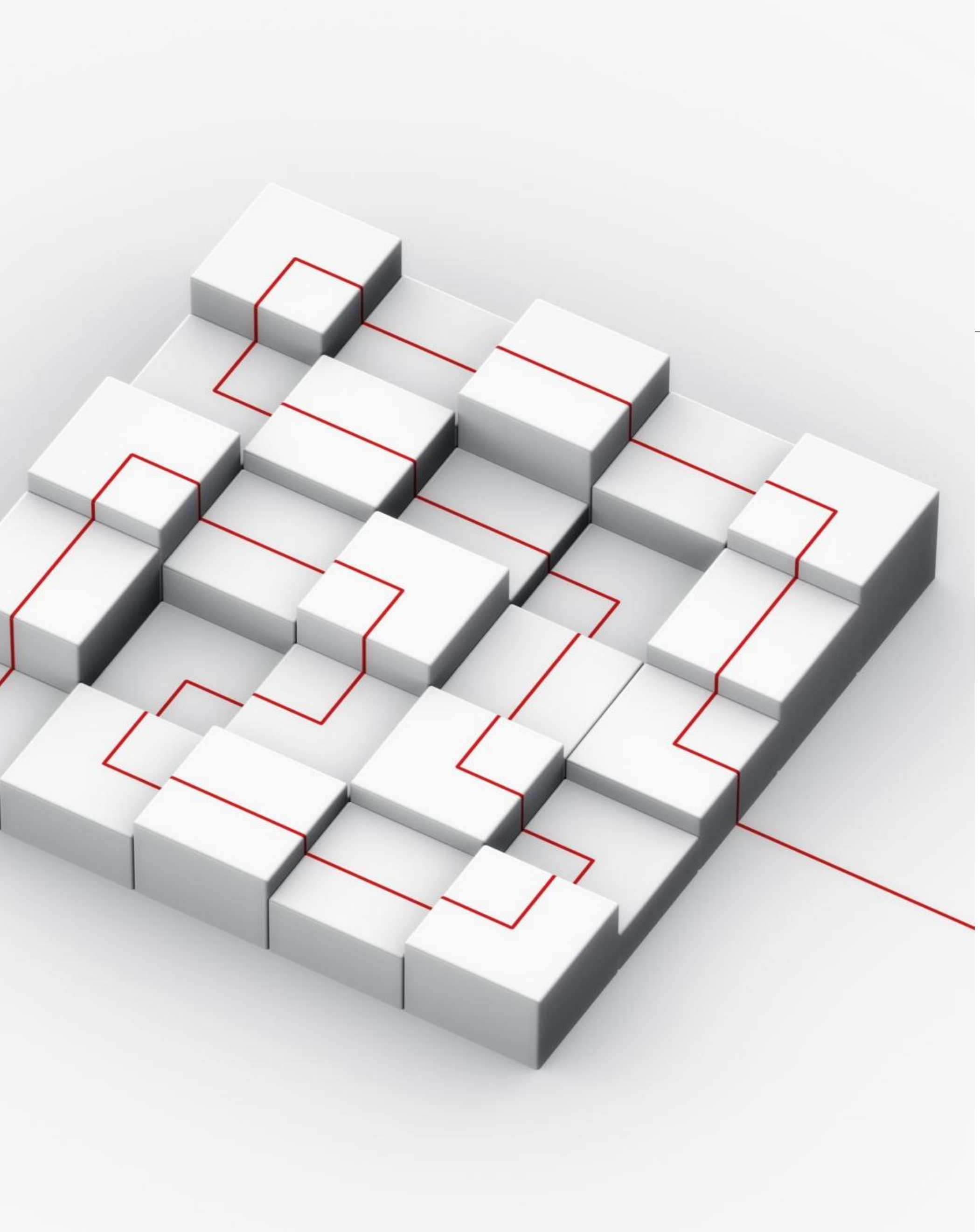
The Orders - The Orders table contains the timestamps for when all table orders were made.

The Order Details - The Order Details table lists the specific pizzas included in each order from the Orders table, along with their quantities.

The Pizzas - The Pizzas table records the size, price, and category for each unique pizza listed in the Order Details table, along with its broader pizza type.

The Pizza Types - The Pizza Types table provides information on the types of pizzas listed in the Pizzas table, including their menu names, categories, and ingredient lists.





DATABASE INFRASTRUCTURE

Our project will utilize MySQL as the database management system, managed through XAMPP. MySQL is a robust and widely-used relational database system known for its performance and scalability. This combination ensures a stable and efficient environment for our data analytics solution at Crispy Crust Corner.

INITIAL DATASET UPLOAD

This slide illustrates the initial step of our data analysis journey, where all four datasets were uploaded into MySQL

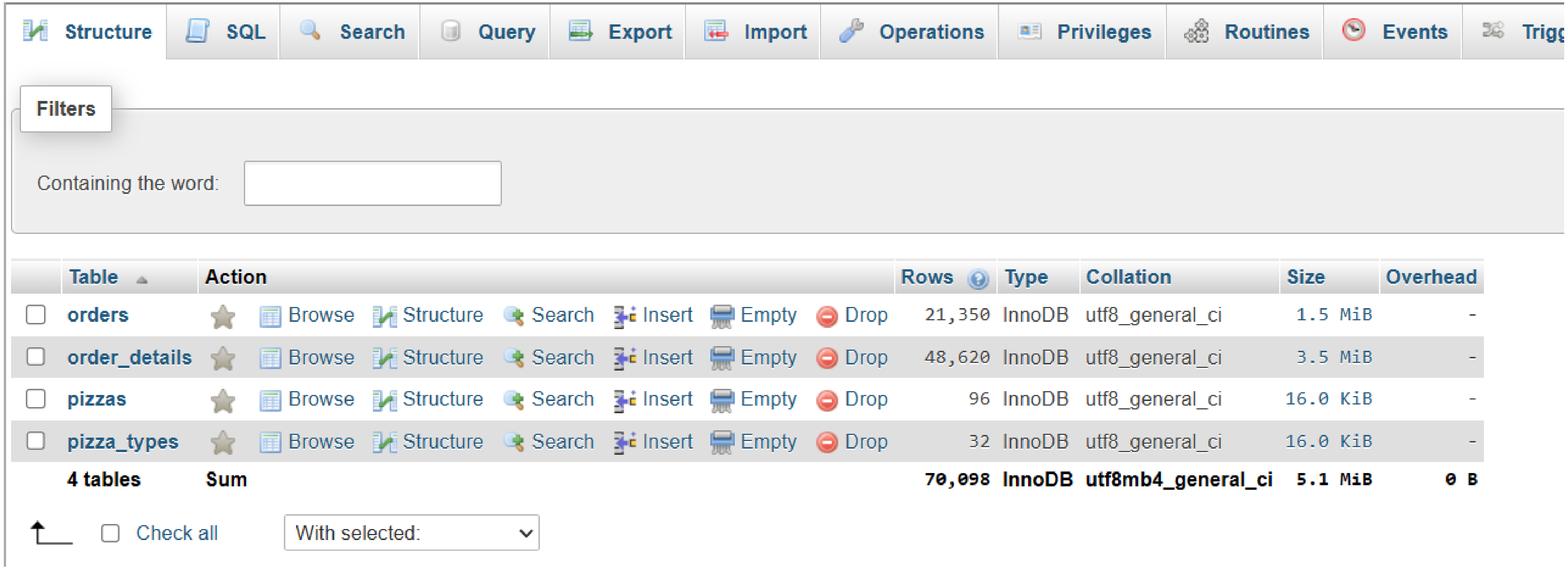


	Table	Action						Rows	Type	Collation	Size	Overhead
	orders		Browse	Structure	Search	Insert	Empty Drop	21,350	InnoDB	utf8_general_ci	1.5 MiB	-
	order_details		Browse	Structure	Search	Insert	Empty Drop	48,620	InnoDB	utf8_general_ci	3.5 MiB	-
	pizzas		Browse	Structure	Search	Insert	Empty Drop	96	InnoDB	utf8_general_ci	16.0 KiB	-
	pizza_types		Browse	Structure	Search	Insert	Empty Drop	32	InnoDB	utf8_general_ci	16.0 KiB	-
	4 tables	**Sum**						**70,098**	**InnoDB**	**utf8mb4_general_ci**	**5.1 MiB**	**0 B**

MySQL DATA CONSOLIDATION

After uploading all four datasets into MySQL, we executed a query to consolidate the data into a single CSV file containing the variables required for the project analysis.

Structure | SQL | Search | Query | Export | Import

Run SQL query/queries on database dummy_project:

```sql
CREATE TABLE pizza_sales AS
SELECT
    od.order_details_id,
    od.order_id,
    od.pizza_id AS order_pizza_id,
    od.quantity,
    o.`date`,
    o.`time`,
    p.pizza_id,
    p.pizza_type_id,
    p.`size`,
    p.price,
    pt.`name`,
    pt.category,
    pt.ingredients
FROM order_details od
LEFT JOIN orders o ON od.order_id = o.order_id
LEFT JOIN pizzas p ON od.pizza_id = p.pizza_id
LEFT JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id;
```

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 61.7846 seconds.)

```sql
CREATE TABLE pizza_sales AS SELECT od.order_details_id,
od.order_id, od.pizza_id AS order_pizza_id, od.quantity,
o.`date`, o.`time`, p.pizza_id, p.pizza_type_id, p.`size`,
p.price, pt.`name`, pt.category, pt.ingredients FROM
order_details od LEFT JOIN orders o ON od.order_id =
o.order_id LEFT JOIN pizzas p ON od.pizza_id = p.pizza_id
```

[Edit inline] [Edit] [Create PHP code]

MySQL - PROJECT STRUCTURE

Below is the project structure within MySQL, showcasing the imported tables and the newly created 'Pizza_sales' table

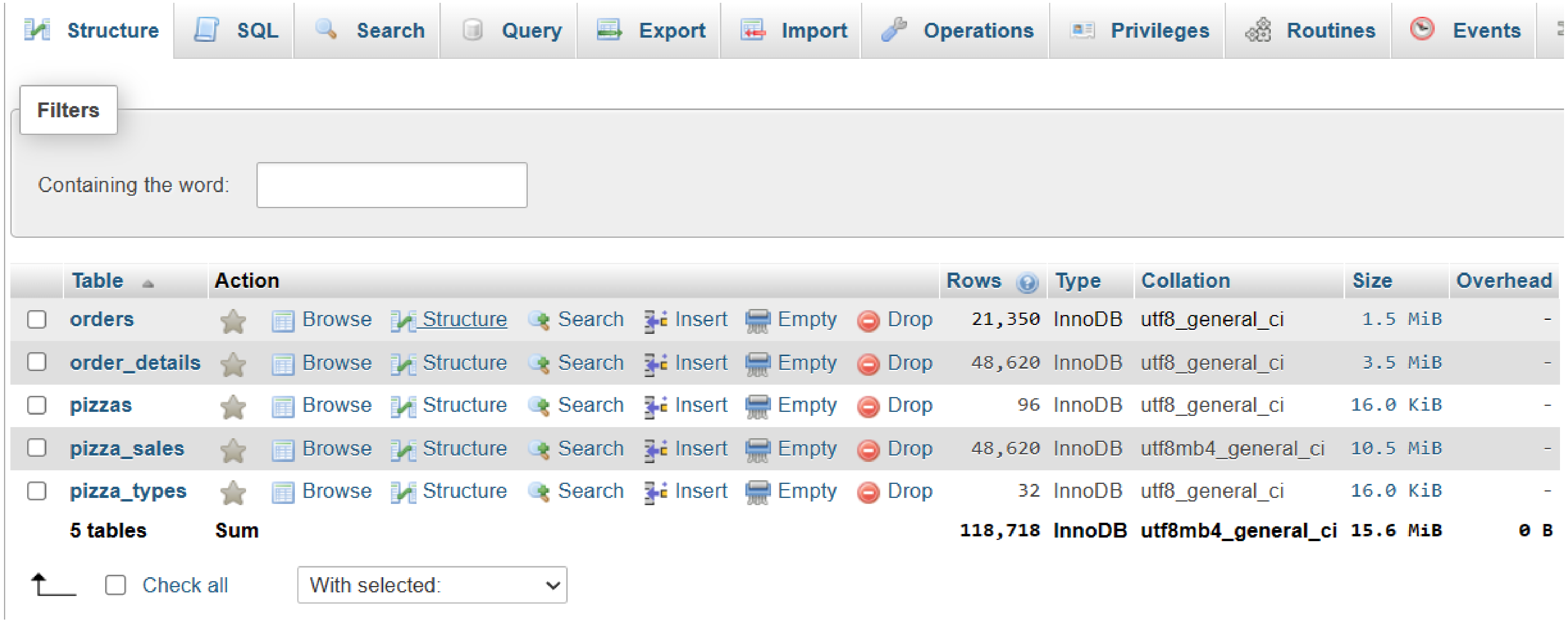


Table	Action						Rows	Type	Collation	Size	Overhead
orders	Browse	Structure	Search	Insert	Empty	Drop	21,350	InnoDB	utf8_general_ci	1.5 MiB	-
order_details	Browse	Structure	Search	Insert	Empty	Drop	48,620	InnoDB	utf8_general_ci	3.5 MiB	-
pizzas	Browse	Structure	Search	Insert	Empty	Drop	96	InnoDB	utf8_general_ci	16.0 KiB	-
pizza_sales	Browse	Structure	Search	Insert	Empty	Drop	48,620	InnoDB	utf8mb4_general_ci	10.5 MiB	-
pizza_types	Browse	Structure	Search	Insert	Empty	Drop	32	InnoDB	utf8_general_ci	16.0 KiB	-
5 tables	Sum						118,718	InnoDB	utf8mb4_general_ci	15.6 MiB	0 B

KEY PERFORMANCE INDICATORS : TOTAL ORDERS

Total Orders Analysis : This slide provides insights into the total number of orders placed at Crispy Crust Corner. Total orders reflect customer engagement and demand for products. Understanding this metric is crucial for assessing business activity and planning operations effectively.

Your SQL query has been executed successfully.

```
SELECT COUNT(DISTINCT order_id) AS total_orders FROM pizza_sales;
```

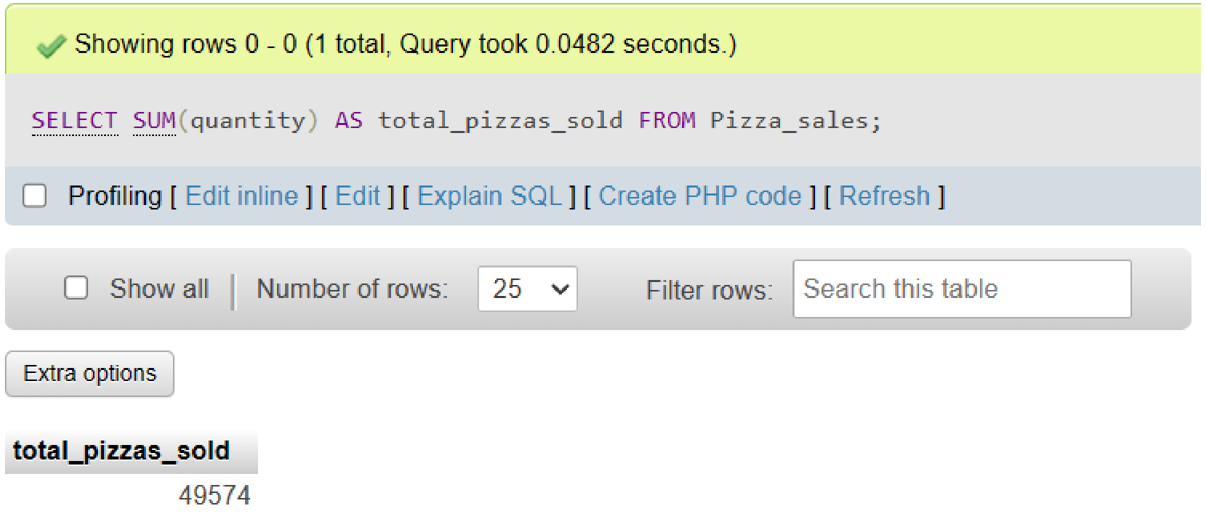
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

total_orders
21350

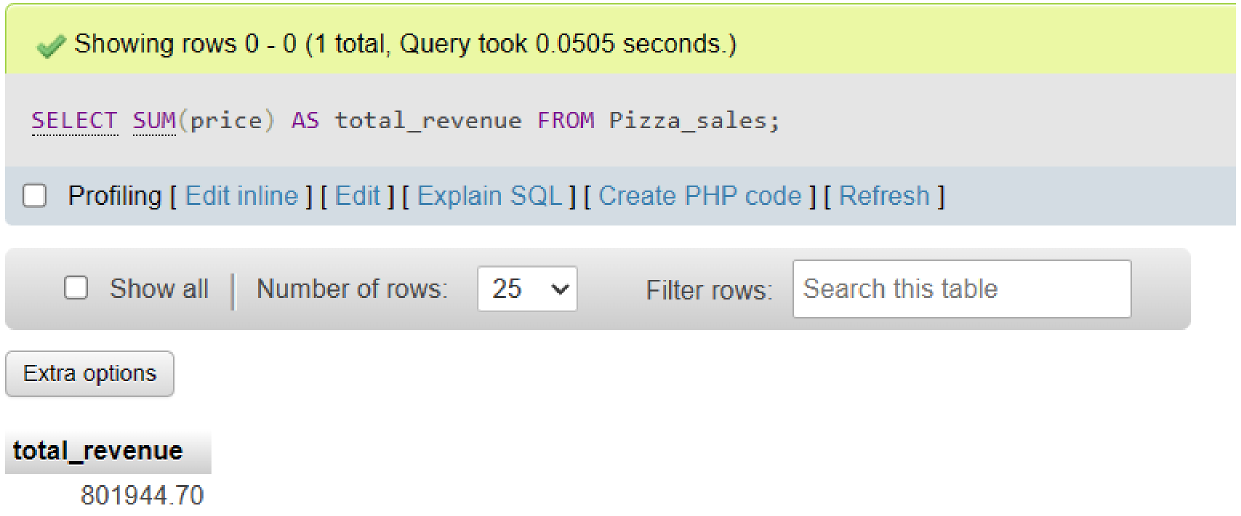
KEY PERFORMANCE INDICATORS : TOTAL PIZZAS SOLD

This slide presents the analysis of total pizzas sold at Crispy Crust Corner. Total pizzas sold reflects product demand and consumption patterns. Understanding this metric aids in inventory management and production planning, ensuring efficient fulfillment of customer demand. It serves as a fundamental indicator of business performance and market demand.



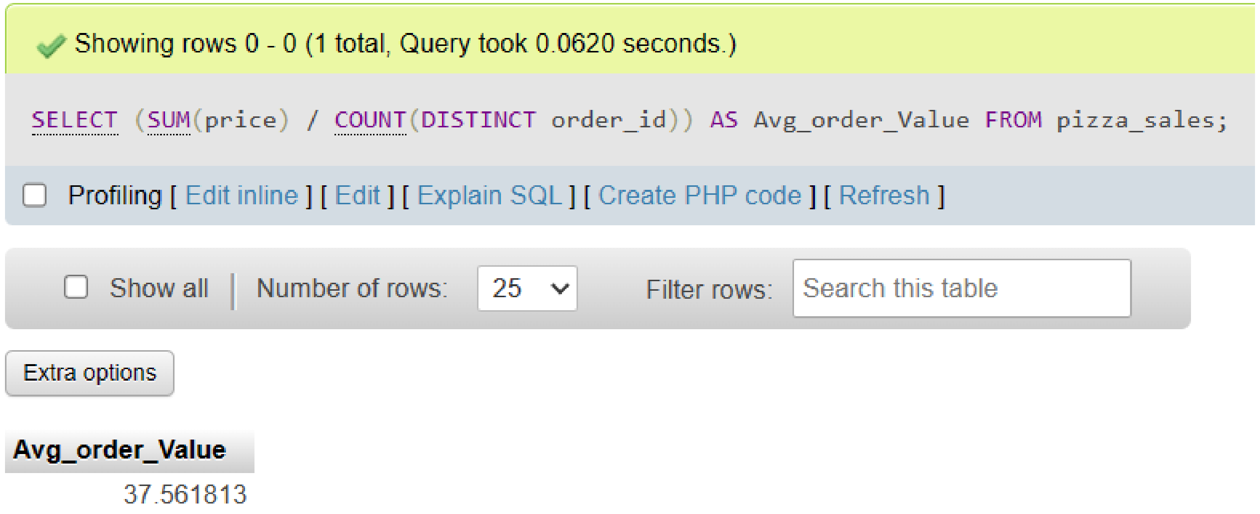
KEY PERFORMANCE INDICATORS : TOTAL REVENUE

This slide presents the analysis of total revenue generated from pizza sales at Crispy Crust Corner. The total revenue is a crucial metric that reflects the financial performance of the business and its sales effectiveness. In the context of our analysis, total revenue serves as a fundamental indicator of Crispy Crust Corner's financial health and success in meeting customer demand



KEY PERFORMANCE INDICATORS : AVERAGE ORDER VALUE

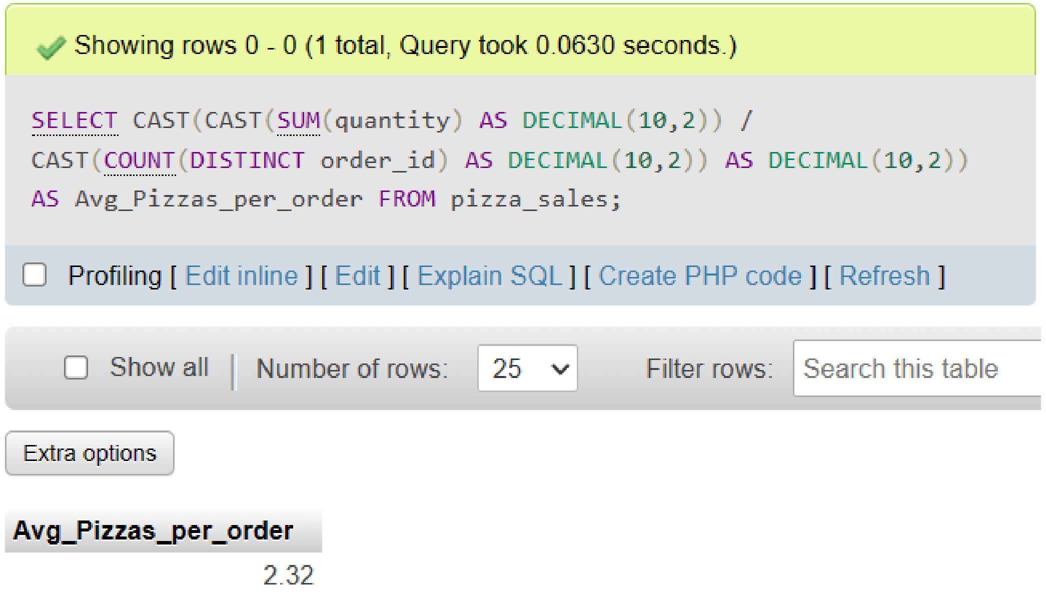
This slide analyzes the average order value (AOV) at Crispy Crust Corner. AOV offers insights into customer spending behavior and pricing effectiveness. AOV indicates typical customer spending per transaction. Monitoring AOV helps optimize pricing strategies and identify upselling opportunities. It serves as a key metric for assessing revenue potential and customer purchasing trends.



KEY PERFORMANCE INDICATORS : AVERAGE PIZZAS PER ORDER

This slide presents the analysis of the average number of pizzas per order at Crispy Crust Corner.

This metric provides insights into customer ordering behavior and preferences, aiding in menu optimization and demand forecasting.



KEY PERFORMANCE INDICATORS : DAILY TREND FOR ORDERS

Daily Trend for orders

Understanding these trends enables businesses to anticipate demand fluctuations and tailor strategies accordingly.

Showing rows 0 - 6 (7 total, Query took 0.1549 seconds.)

```
SELECT DAYNAME(date) AS order_day, COUNT(DISTINCT
order_id) AS total_orders FROM pizza_sales GROUP BY
DAYNAME(date) ORDER BY total_orders DESC;
```

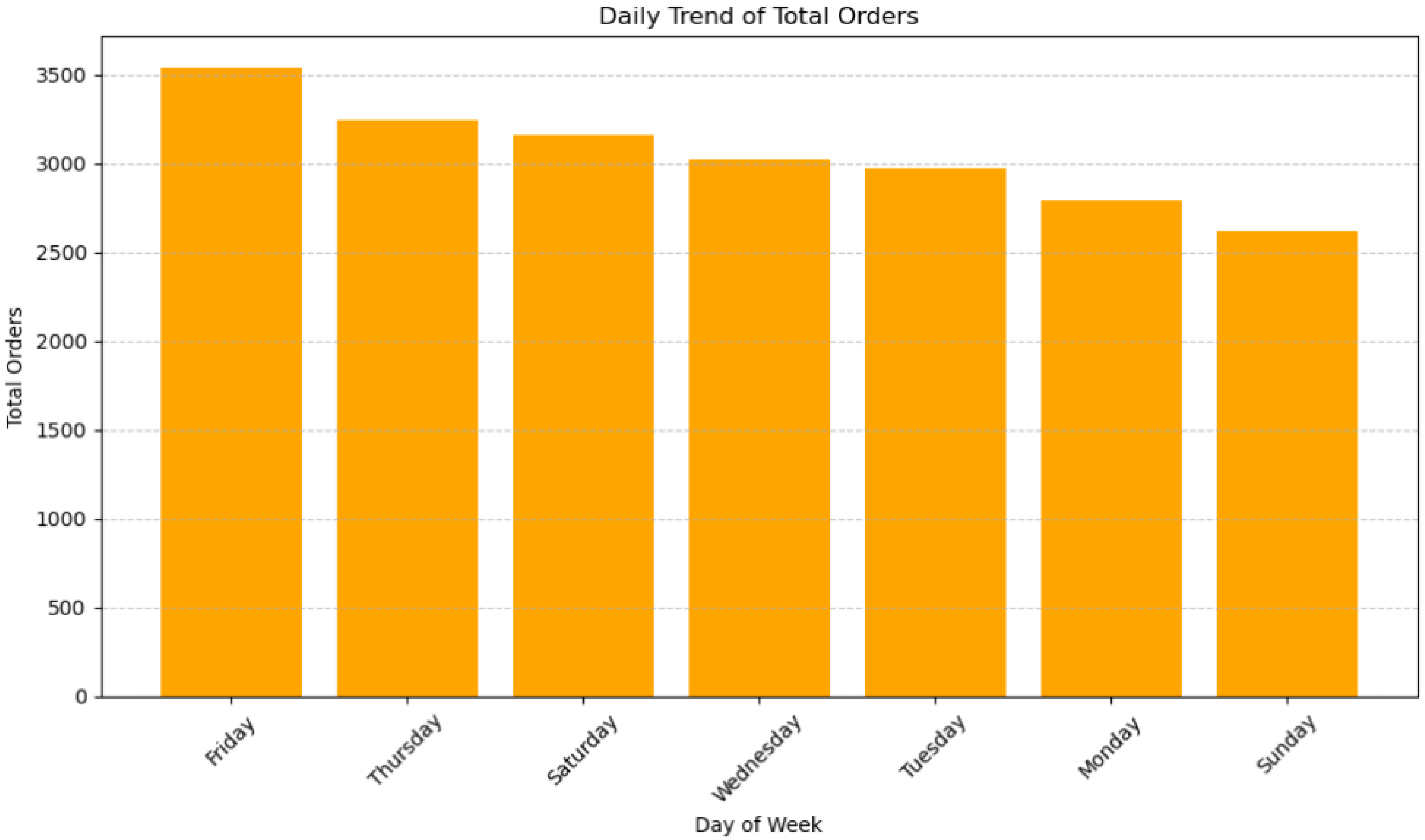
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows:

Extra options

order_day	total_orders 1
Friday	3538
Thursday	3239
Saturday	3158
Wednesday	3024
Tuesday	2973
Monday	2794
Sunday	2624

KEY PERFORMANCE INDICATORS : DAILY TREND FOR ORDERS



KEY PERFORMANCE INDICATORS : MONTHLY TREND FOR ORDERS

Monthly Trend for orders

Understanding these trends enables businesses to anticipate demand fluctuations and tailor strategies accordingly.

Showing rows 0 - 11 (12 total, Query took 0.1142 seconds.)

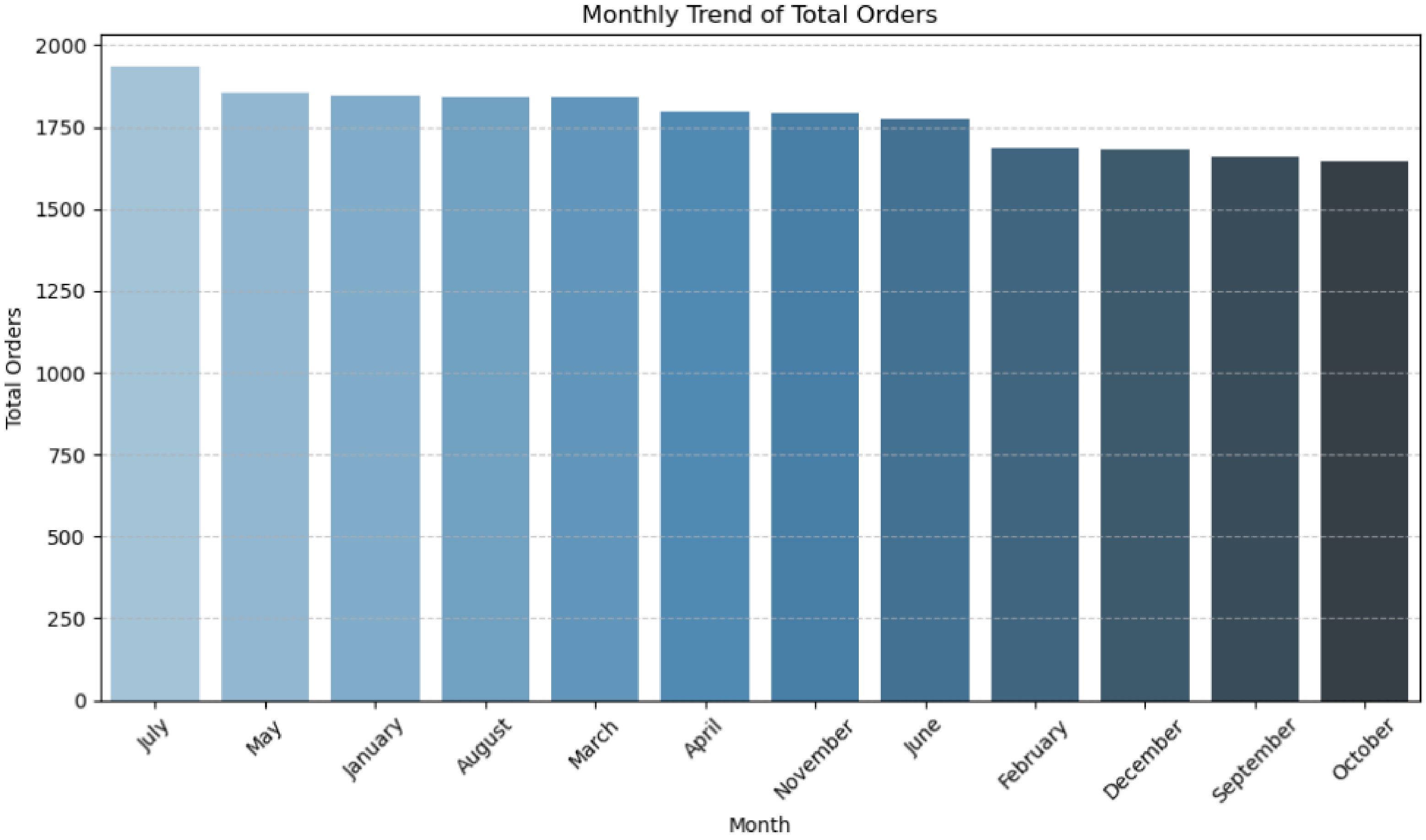
```
SELECT MONTHNAME(date) AS Month_Name, COUNT(DISTINCT order_id)
AS Total_Orders FROM pizza_sales GROUP BY MONTHNAME(date) LIMIT
0, 25;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

Month_Name	Total_Orders
April	1799
August	1841
December	1680
February	1685
January	1845
July	1935
June	1773
March	1840
May	1853
November	1792
October	1646
September	1661

KEY PERFORMANCE INDICATORS : MONTHLY TREND FOR ORDERS



KEY PERFORMANCE INDICATORS : PERCENTAGE OF SALES BY PIZZA CATEGORY

This slide presents the analysis of sales distribution across pizza categories at Crispy Crust Corner. It highlights the total revenue generated by each category and the percentage of sales it contributes relative to the total revenue.

Understanding these percentages aids in identifying top-performing categories and optimizing menu offerings to meet customer preferences and maximize revenue.

Showing rows 0 - 3 (4 total, Query took 0.1219 seconds.)

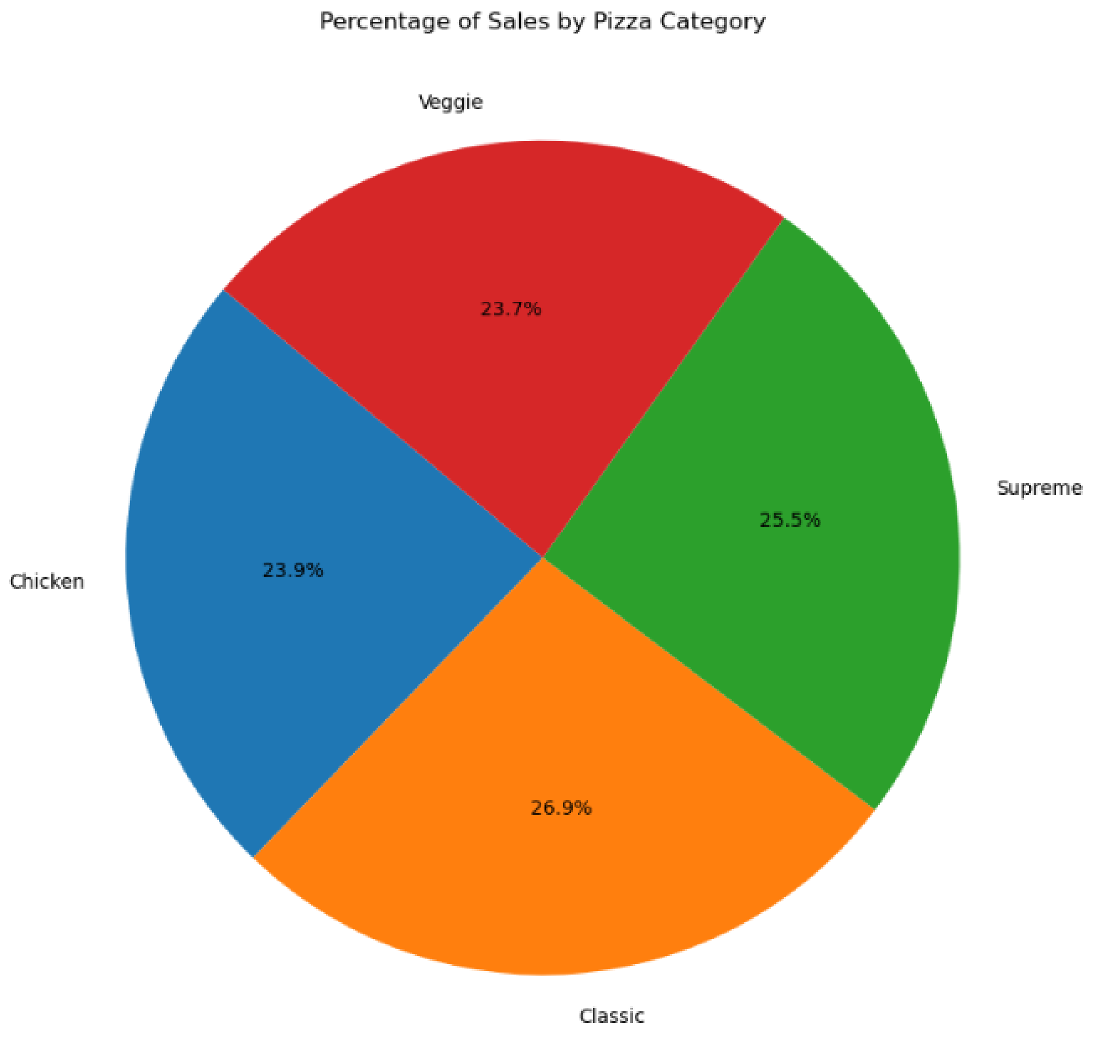
```
SELECT category, CAST(SUM(price) AS DECIMAL(10,2)) as
total_revenue, CAST(SUM(price) * 100 / (SELECT SUM(price)
from pizza_sales) AS DECIMAL(10,2)) AS PCT FROM pizza_sales
GROUP BY category;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this

Extra options

category	total_revenue	PCT
Chicken	191527.25	23.88
Classic	215732.60	26.90
Supreme	204486.75	25.50
Veggie	190198.10	23.72



PERCENTAGE OF SALES BY PIZZA CATEGORY

KEY PERFORMANCE INDICATORS : PERCENTAGE OF SALES BY PIZZA SIZE

This slide illustrates the distribution of sales by pizza size at Crispy Crust Corner. It showcases the total revenue generated by each size category and the percentage of sales it contributes relative to the total revenue.

Understanding these percentages aids in optimizing portion sizes and pricing strategies to align with customer preferences and maximize revenue.

Showing rows 0 - 4 (5 total, Query took 0.1134 seconds.)

```
SELECT size, SUM(price) AS total_revenue, (SUM(price) /
(SELECT SUM(price) FROM pizza_sales)) * 100 AS
sales_percentage FROM pizza_sales GROUP BY size;
```

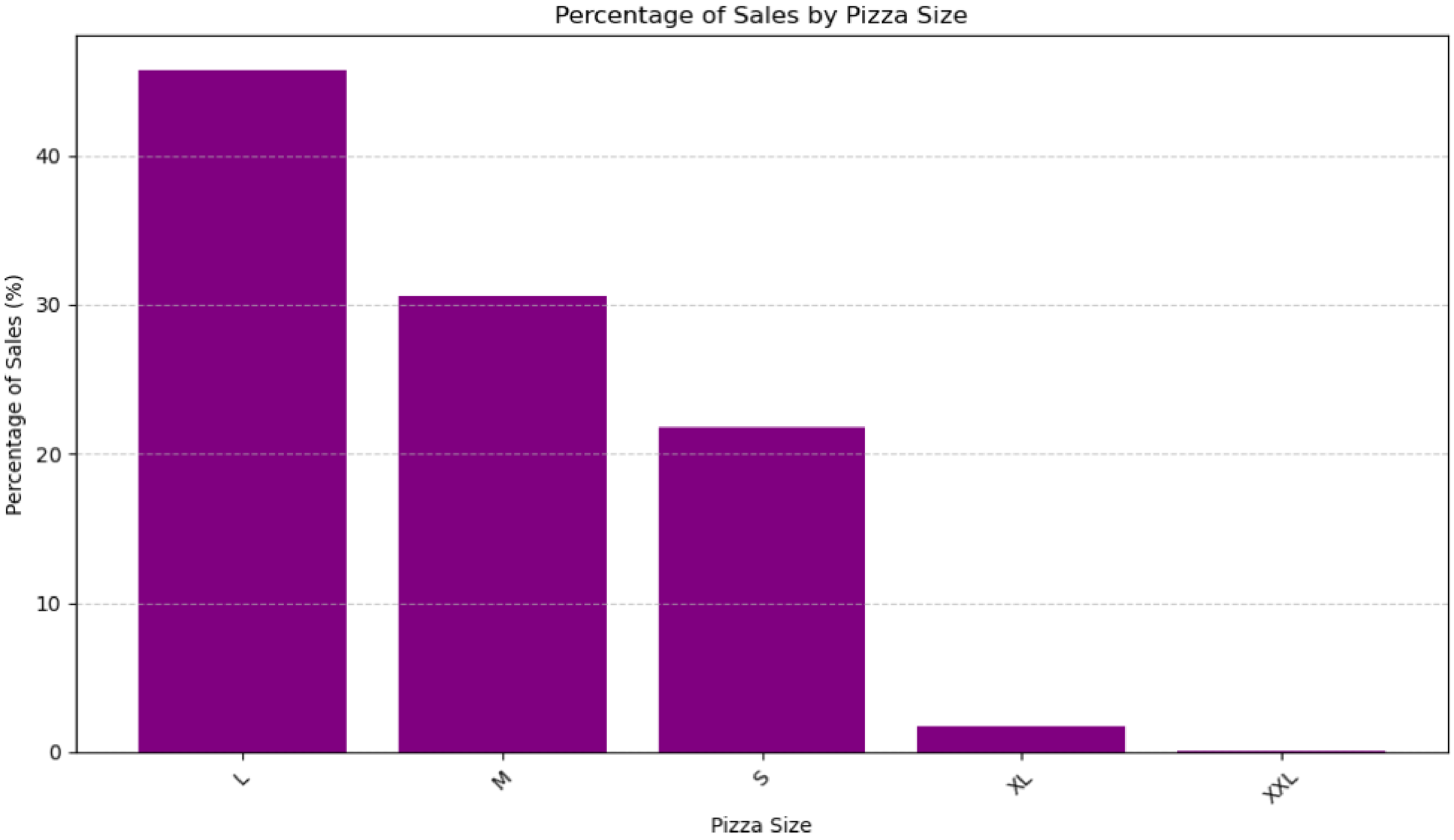
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this ta

Extra options

size	total_revenue	sales_percentage
L	366862.10	45.746558
M	245409.50	30.601798
S	174794.50	21.796328
XL	13872.00	1.729795
XXL	1006.60	0.125520

KEY PERFORMANCE INDICATORS : PERCENTAGE OF SALES BY PIZZA SIZE



KEY PERFORMANCE INDICATORS : TOTAL PIZZAS SOLD BY PIZZA CATEGORY

This slide showcases the total number of pizzas sold categorized by pizza type at Crispy Crust Corner. It provides insights into the popularity of each pizza category and helps in understanding customer preferences.

Understanding these sales figures aids in inventory management and menu optimization to meet customer demand effectively.

Showing rows 0 - 3 (4 total, Query took 0.0837 seconds.)

```sql
SELECT category, SUM(quantity) AS total_pizzas_sold FROM pizza_sales GROUP BY category;
```

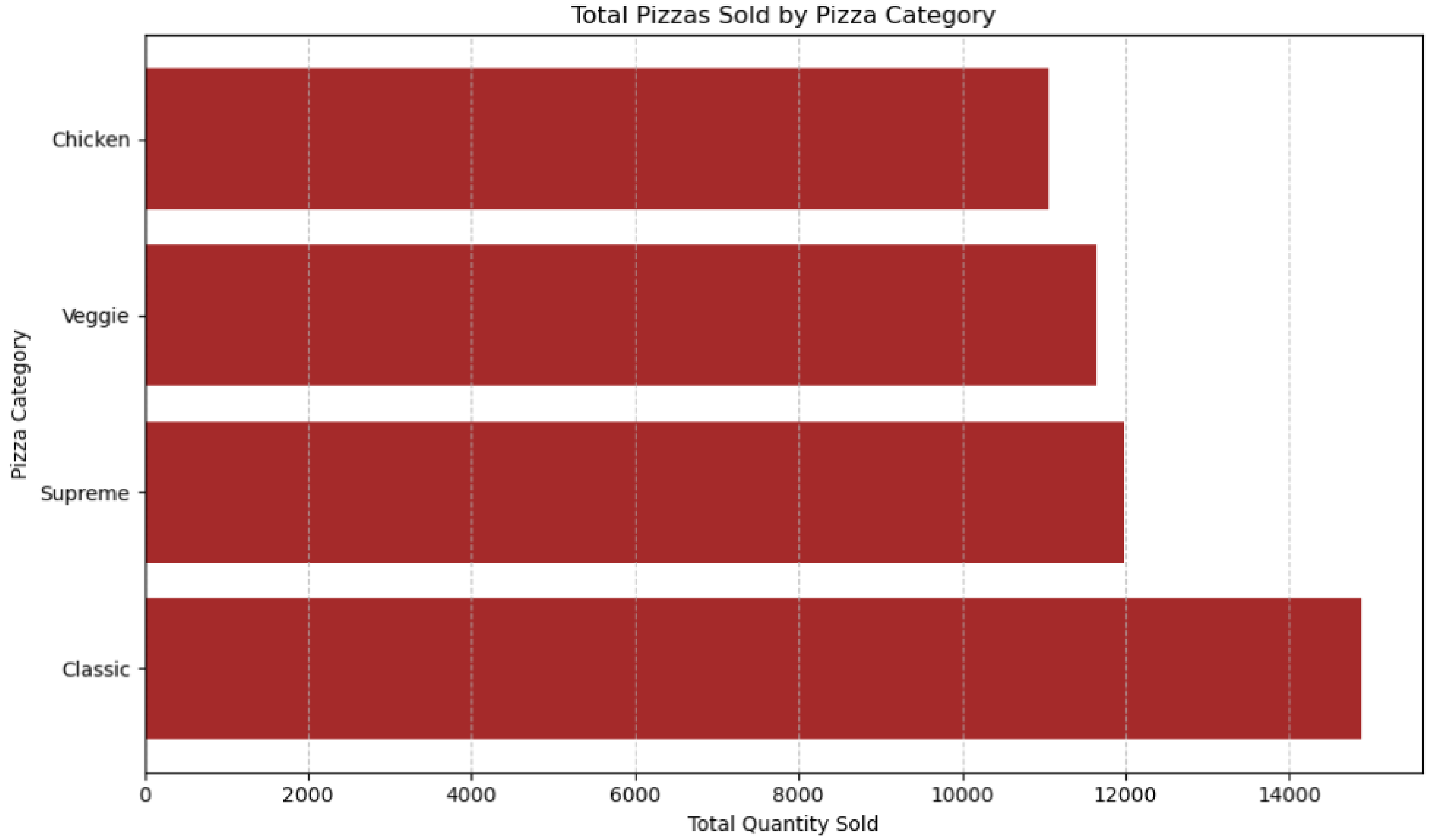
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows:

Extra options

category	total_pizzas_sold
Chicken	11050
Classic	14888
Supreme	11987
Veggie	11649

KEY PERFORMANCE INDICATORS : TOTAL PIZZAS SOLD BY PIZZA CATEGORY



KEY PERFORMANCE INDICATORS : TOP 5 PIZZAS BY QUANTITY

This slide showcases the top 5 pizzas at Crispy Crust Corner based on quantity sold. It provides insights into the most popular menu items in terms of customer demand.

Understanding these top-selling pizzas aids in inventory management and menu optimization to meet customer preferences effectively.

Showing rows 0 - 4 (5 total, Query took 0.1124 seconds.)

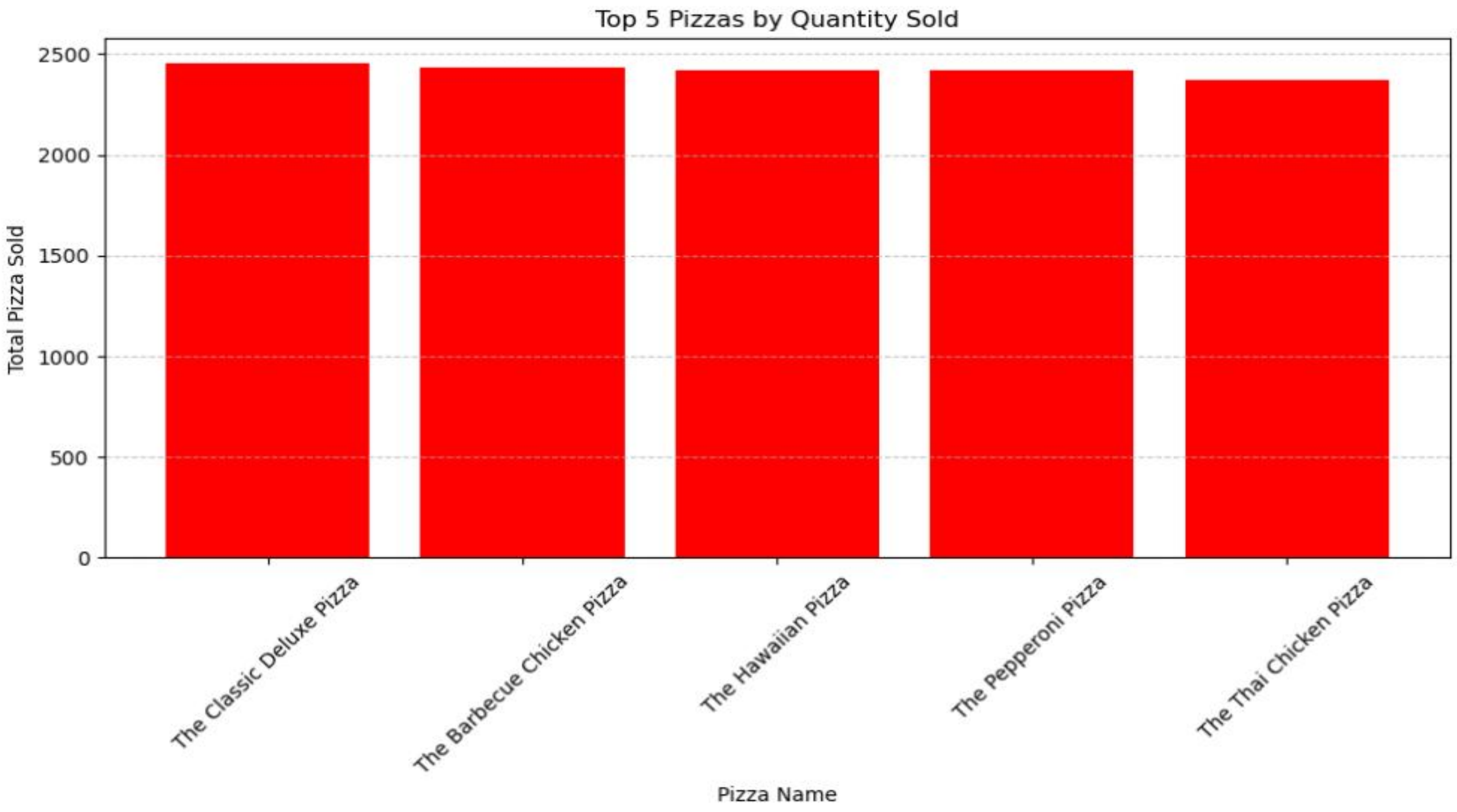
```
SELECT name, SUM(quantity) AS Total_Pizza_Sold FROM
pizza_sales GROUP BY name ORDER BY Total_Pizza_Sold DESC
LIMIT 5;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

name	Total_Pizza_Sold 1
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

KEY PERFORMANCE INDICATORS : TOP 5 PIZZAS BY QUANTITY



KEY PERFORMANCE INDICATORS : TOP 5 PIZZAS BY REVENUE

This slide highlights the top 5 pizzas at Crispy Crust Corner based on revenue generated. It showcases the total revenue earned by each pizza type, providing insights into the most profitable menu items.

Understanding these top-selling pizzas aids in menu optimization and marketing strategies to maximize revenue.

Showing rows 0 - 4 (5 total, Query took 0.1139 seconds.)

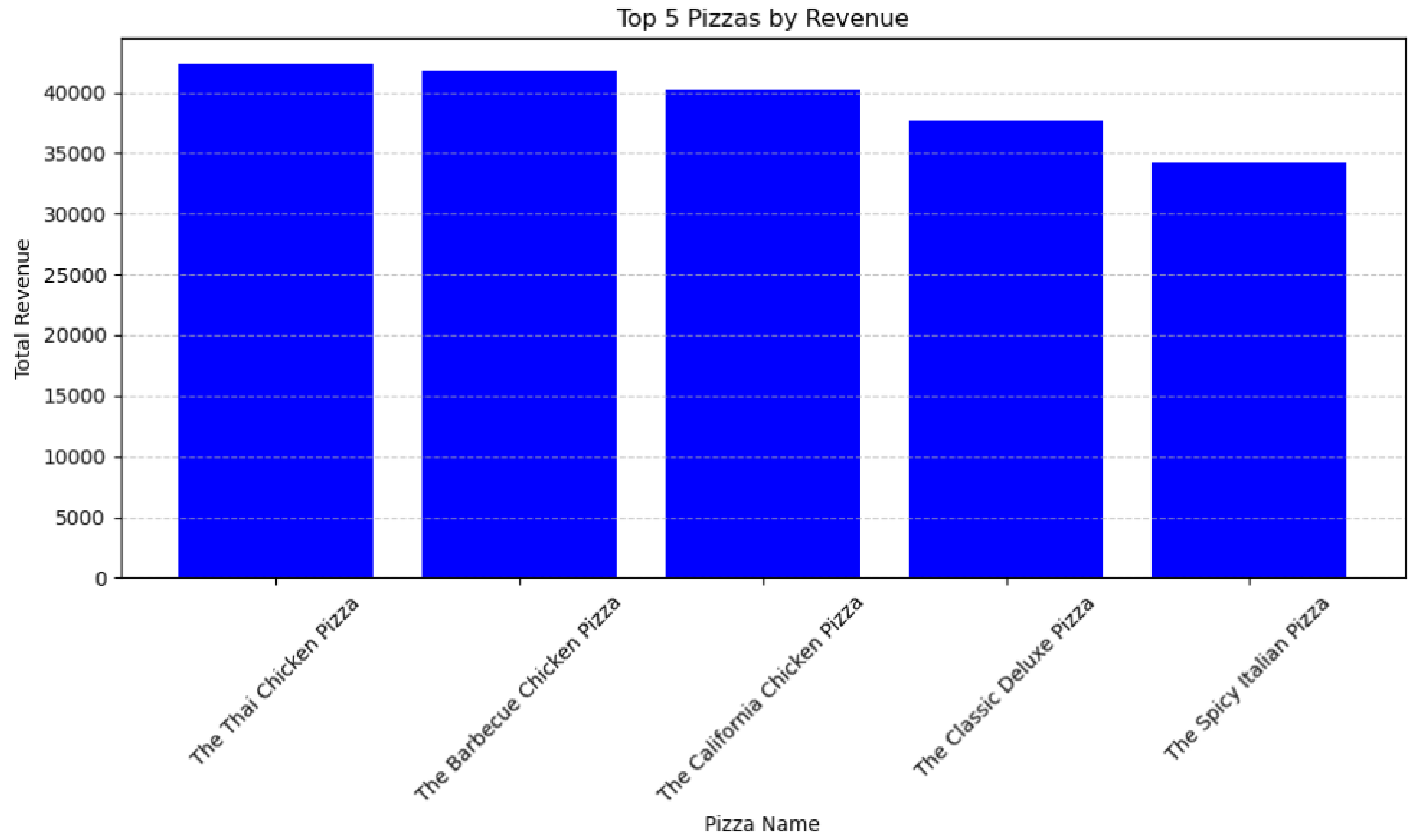
```
SELECT name, SUM(price) AS Total_Revenue FROM pizza_sales
GROUP BY name ORDER BY Total_Revenue DESC LIMIT 5;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

name	Total_Revenue 1
The Thai Chicken Pizza	42332.25
The Barbecue Chicken Pizza	41683.00
The California Chicken Pizza	40166.50
The Classic Deluxe Pizza	37631.50
The Spicy Italian Pizza	34163.50

KEY PERFORMANCE INDICATORS : TOP 5 PIZZAS BY REVENUE



KEY PERFORMANCE INDICATORS : BOTTOM 5 PIZZAS BY QUANTITY

This slide highlights the bottom 5 pizzas at Crispy Crust Corner based on quantity sold. It provides insights into less popular menu items in terms of customer demand.

Understanding these bottom-selling pizzas aids in inventory management and menu optimization to improve sales.

Showing rows 0 - 4 (5 total, Query took 0.1141 seconds.)

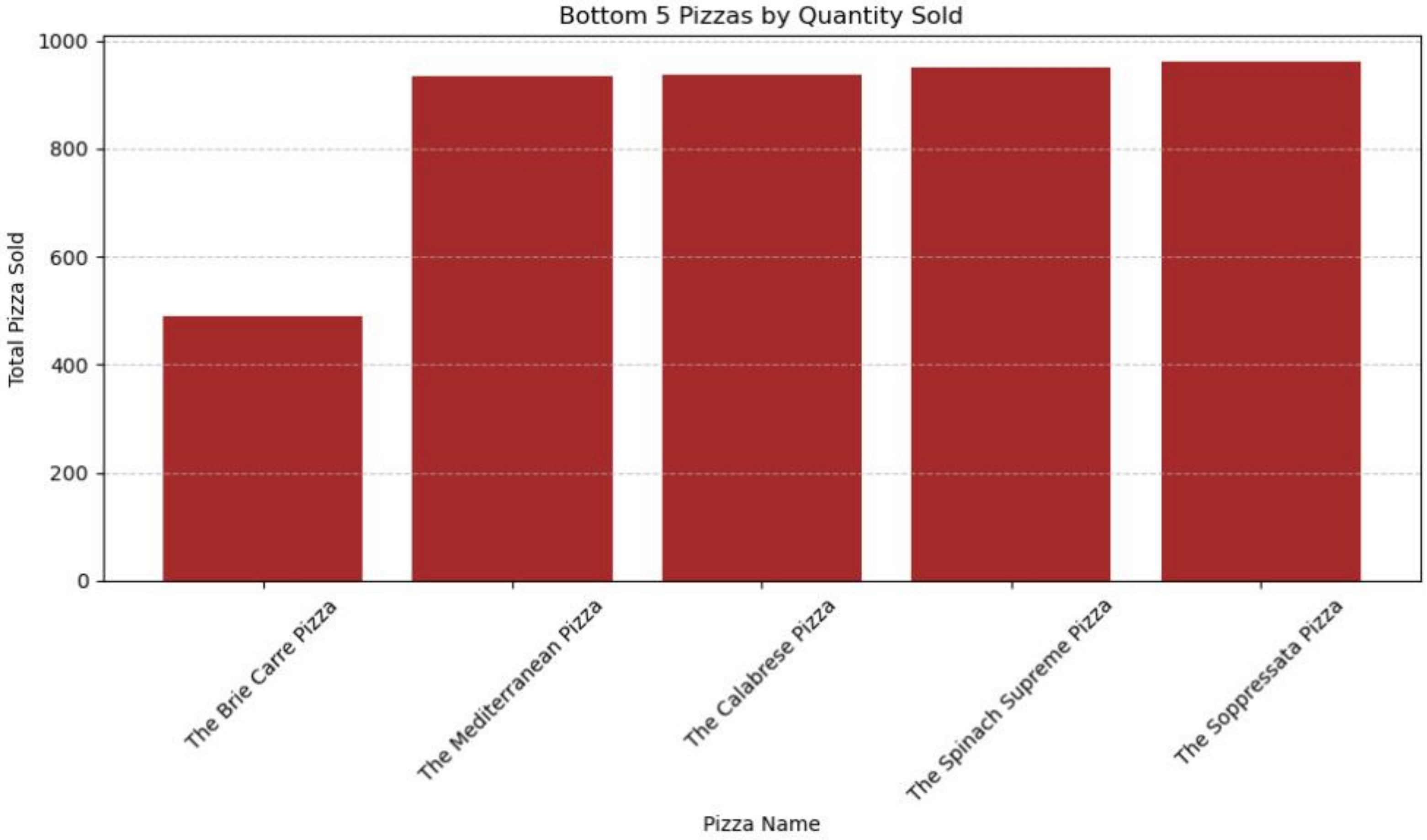
```
SELECT name, SUM(quantity) AS Total_Pizza_Sold FROM
pizza_sales GROUP BY name ORDER BY Total_Pizza_Sold ASC LIMIT
5;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

name	Total_Pizza_Sold 1
The Brie Carre Pizza	490
The Mediterranean Pizza	934
The Calabrese Pizza	937
The Spinach Supreme Pizza	950
The Soppressata Pizza	961

KEY PERFORMANCE INDICATORS : BOTTOM 5 PIZZAS BY QUANTITY



KEY PERFORMANCE INDICATORS : BOTTOM 5 PIZZAS BY REVENUE

This slide highlights the bottom 5 pizzas at Crispy Crust Corner based on revenue generated. It showcases the total revenue earned by each pizza type, providing insights into less profitable menu items.

Understanding these bottom-selling pizzas aids in identifying opportunities for menu adjustments and marketing strategies to improve revenue.

Showing rows 0 - 4 (5 total, Query took 0.1140 seconds.)

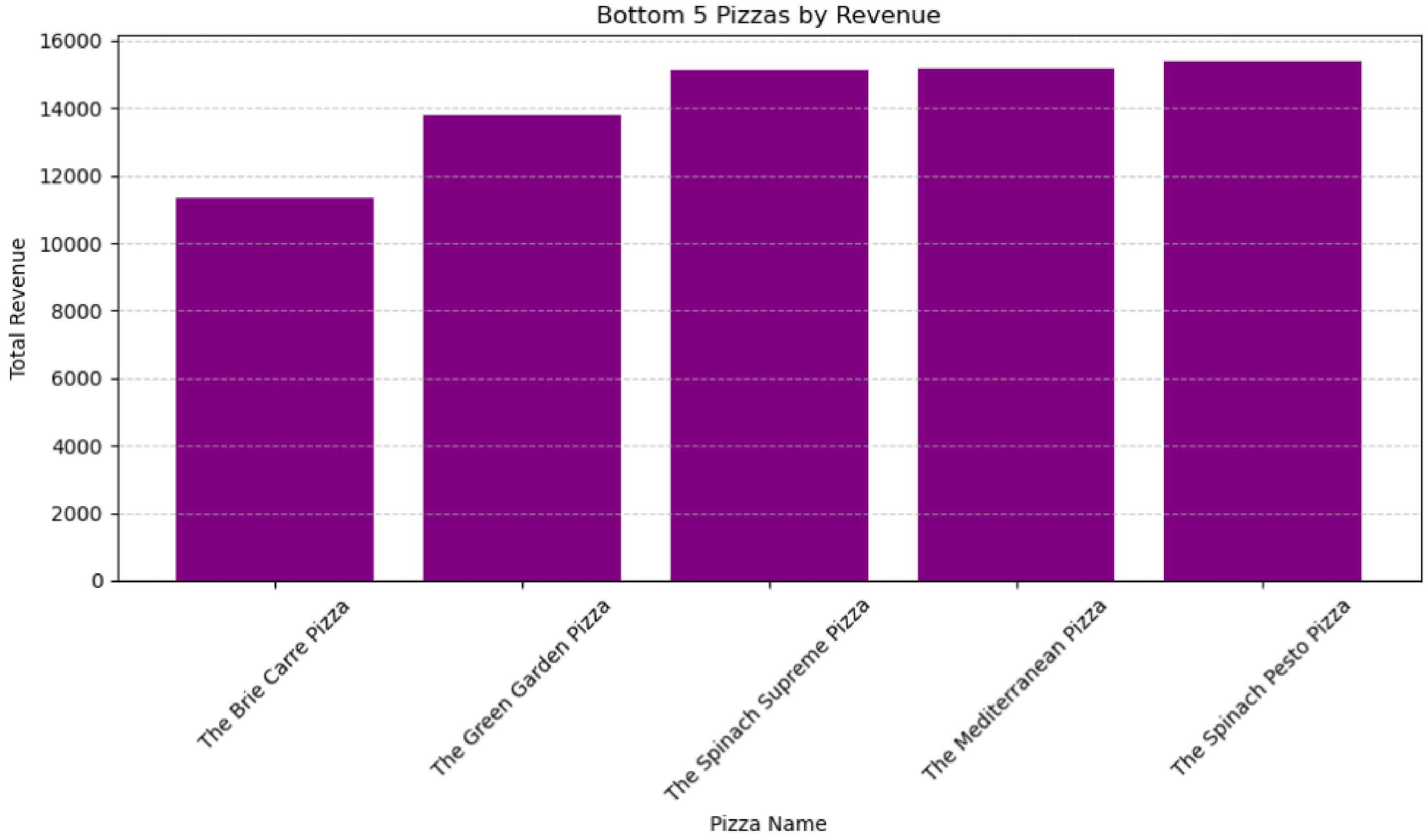
```
SELECT name, SUM(price) AS Total_Revenue FROM pizza_sales
GROUP BY name ORDER BY Total_Revenue ASC LIMIT 5;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

name	Total_Revenue 1
The Brie Carre Pizza	11352.00
The Green Garden Pizza	13819.50
The Spinach Supreme Pizza	15124.00
The Mediterranean Pizza	15163.00
The Spinach Pesto Pizza	15388.25

KEY PERFORMANCE INDICATORS : BOTTOM 5 PIZZAS BY REVENUE



CONCLUSION : DATA-DRIVEN INSIGHTS

Based on the comprehensive analysis conducted and the provided key performance indicators, it's evident that Crispy Crust Corner can leverage data analytics to gain valuable insights into their business operations.

By implementing a robust and scalable data analytics solution, Crispy Crust Corner can efficiently process large volumes of data to uncover trends, ordering behaviors, menu performance, customer preferences, and essential business metrics.

These insights enable informed decision-making, allowing Crispy Crust Corner to optimize menu offerings, pricing strategies, inventory management, and marketing efforts, ultimately driving business growth and enhancing customer satisfaction.

Through the integration of data analytics into their operations, Crispy Crust Corner can well-position itself to remain competitive in the dynamic pizza industry landscape.

THANK YOU!